Saroj Khadka

Graduate Researcher

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Summary

Biomedical Science PhD candidate with strong research experience in bacterial virulence mechanisms, host-pathogen interactions, and microbial gene regulation. Skilled in molecular biology and microbiological techniques, mentoring and science communication. Motivated to build an independent career in biomedical research.

Education

University of Pittsburgh, Microbiology and Immunology — Pittsburgh, PA, USA — 2024 – 2026
- Description 1.
- Description 2.

Tribhuvan University, Medical Microbiology — Kathmandu, Nepal — 2017 – 2019
- Description 1.
- Description 2.

Experience

Division of Infectious Diseases/Department of Medicine, University of Pittsburgh, Graduate Student Researcher — University of Pittsburgh, PA — 2024 – 2026 — 2 years

Teaching at Palmer Physical Laboratory (now 302 Frist Campus Center). While not a professor at Princeton, I associated with the physics professors and continued to give lectures on campus.
- Relativity
- Description 2.

B-Sure Path Lab & Diagnostic Centre, Molecular Biologist — Biratnagar, Nepal — 2020 – 2021 — 1 year
- Description 1.
- Description 2.

Volunteer

People's Climate March, Lead Organizer — Zurich, Switzerland — Apr 2014 – July 2015

Lead organizer for the New York City branch of the People's Climate March, the largest climate march in history.
- Awarded 'Climate Hero' award by Greenpeace for my efforts organizing the march.
- Men of the year 2014 by Time magazine

Awards

AHA Predoctoral Fellow — Jan 2022

The Nobel Prizes are five separate prizes that, according to Alfred Nobel's will of 1895, are awarded to 'those who, during the preceding year, have conferred the greatest benefit to humankind.'

American Heart Association

Max Planck Medal — 2029

Awarded for outstanding scientific achievement

German Physical Society

Publications

Zur Elektrodynamik bewegter Körper
It concerned an interpretation of the Michelson–Morley experiment and the properties of light and time. Special relativity incorporates the principle that the speed of light is the same for all inertial observers regardless of the state of motion of the source.
Albert Einstein
en.wikisource.org/wiki/Translation:On_the_Electrodynamics_of_Moving_Bodies

Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt
In the second paper, he applied the quantum theory to light to explain the photoelectric effect. In particular, he used the idea of light quanta (photons) to explain experimental results, but stressed the importance of the experimental results. The importance of his work on the photoelectric effect earned him the Nobel Prize in Physics in 1921.
Albert Einstein
de.wikisource.org/wiki/%C3%9Cber_einen_die_Erzeugung_und_Verwandlung_des_Lichtes_betreffenden_heuristischen_Gesichtspun

Die Grundlage der allgemeinen Relativitätstheorie
The publication of the theory of general relativity made him internationally famous. He was professor of physics at the universities of Zurich (1909–1911) and Prague (1911–1912), before he returned to ETH Zurich (1912–1914).
Albert Einstein
de.wikisource.org/wiki/Die_Grundlage_der_allgemeinen_Relativit%C3%A4tstheorie

Skills

Physics

Languages

German
Native speaker

English
Fluent

Interests

Physics

Certificates

Machine Learning Jan 2018

Quantum Computing Jan 2018

Quantum Information Jan 2018

Projects

Quantum Computing Jan 2018 – Jan 2018
Quantum computing is the use of quantum-mechanical phenomena such as superposition and entanglement to perform computation. Computers that perform quantum computations are known as quantum computers.

- Quantum Teleportation
- Quantum Cryptography

References

Professor John Doe

Professor Jane Smith